MMP 1
Serie 5
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1 MC

1.a

(C)

2

2.a

Since $f, g \in L^1(\mathbb{R}/\mathbb{Z})$ $f(x) = f^{\wedge\vee}$

$$\begin{aligned}(f \star g)(x) &= \int_{\mathbb{R}} f(y)g(x-y)dy \\ &= \int_{\mathbb{R}} f^{\wedge\vee}(y)g(x-y)dy \\ &= \int_{\mathbb{R}} \hat{f}(y)\hat{g}(x-y)dy \\ &= \left(\hat{f} \star \hat{g}\right)(x)\end{aligned} \tag{2.1}$$

2.b

$$\begin{aligned}\widehat{\left(f \star g\right)}(k) &= \int_{\mathbb{R}^n} e^{-ikx} \int_{\mathbb{R}^n} g(y)f(x-y)dydx \\ &\underset{\text{fubini}}{=\!=} \int_{\mathbb{R}^n} g(y) \int_{\mathbb{R}^n} f(x-y)e^{-ikx}dxdy \\ &= \int_{\mathbb{R}^n} g(y) \int_{\mathbb{R}^n} f(z)e^{-ikz}e^{-iky}dzdy \\ &= \hat{f}(k) \int_{\mathbb{R}^n} g(y)e^{-iky}dy = \hat{f}(k)\hat{g}(k)\end{aligned} \tag{2.2}$$

2.c

$$f = \begin{cases} 0 & |x| \geq 1 \\ 1 - |x| & \text{else} \end{cases} \tag{2.3}$$

$$\begin{aligned}\hat{f}(k) &= \int_{-1}^{0} (1+x)e^{-ikx}dx + \int_{0}^{1} (1-x)e^{-ikx}dx \\ &= \left[\frac{1+x}{-ik}e^{-ikx}\right]_{-1}^{0} + \left[\frac{1-x}{-ik}e^{-ikx}\right]_{0}^{1} + \int_{-1}^{0} \frac{1}{ik}e^{-ikx}dx - \int_{0}^{1} \frac{1}{ik}e^{-ikx}dx \\ &= \frac{i}{k} - \frac{i}{k} + \left[\frac{1}{k^2}e^{-ikx}\right]_{-1}^{0} + \left[-\frac{1}{k^2}e^{-ikx}\right]_{0}^{1} \\ &= \frac{1}{k^2} - \frac{e^{ik}}{k^2} - \frac{e^{-ik}}{k^2} + \frac{1}{k^2} = \frac{2}{k^2}(1-\cos(k))\end{aligned} \tag{2.4}$$

$k = 0$

$$\begin{aligned}\hat{f}(0) &= \int_{-1}^{0} 1 + xdx + \int_{0}^{1} 1 - xdx \\ &= \frac{1}{2} + \frac{1}{2} = 1\end{aligned} \tag{2.5}$$

3

3.a

$\tau = 2\pi i$

$$f(x) = \min\{x, 1-x\}, \qquad x \in [0,1) \tag{3.1}$$

$$
\begin{aligned}
f_n &= \int_0^1 f(x)e^{-\tau nx}dx \\
&= \int_0^{\frac{1}{2}} (1-x)e^{-\tau nx}dx + \int_{\frac{1}{2}}^1 xe^{-\tau nx}dx \\
&= \left[\frac{e^{-tnx}}{-\tau n}(1-x)\right]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} -\frac{1}{-\tau n}e^{-\tau nx}dx + \left[\frac{x}{-\tau n}e^{-\tau nx}\right]_{\frac{1}{2}}^1 - \int_{\frac{1}{2}}^1 \frac{1}{-\tau n}e^{-\tau nx}dx \qquad (3.2)\\
&= \frac{1+\overbrace{e^{2\pi in/2}-e^{-2\pi in/2}}^{=0}}{4\pi in} - \frac{e^{-\tau n}}{\tau n} + \left[\frac{1}{\tau^2 n^2}e^{-\tau nx}\right]_0^{\frac{1}{2}} - \left[\frac{1}{\tau^2 n^2}e^{-\tau nx}\right]_{\frac{1}{2}}^1 \\
&= \frac{2e^{-\tau n/2}-1-e^{-\tau n}}{\tau^2 n^2} = \frac{2((-1)^n-1)}{\tau^2 n^2}
\end{aligned}
$$

$n = 0$

$$f_0 = \frac{1}{4} \qquad (3.3)$$

habe es für max gemacht :(
KI ⇒

$$\Rightarrow f_n = \frac{(-1)^2-1}{2\pi^2 n^2} \qquad (3.4)$$

3.b

$$
\begin{aligned}
(u_x)_{n,m} &= \int_{[0,1]\times[0,1]} u_x(x,t)e^{-2\pi i(nx+my)}dxdx \\
&= \underbrace{\left[\frac{u(x,t)}{-2\pi in}e^{-2\pi i(nx+my)}\right]_{[0,1]\times[0,1]}}_{} - \int_{[0,1]\times[0,1]} (-2\pi in)u(x,t)e^{-2\pi i(nx+my)}dxdy \qquad (3.5)\\
&= 2\pi i n u_{n,m}
\end{aligned}
$$

$$\Rightarrow \sum_{\substack{k\in\mathbb{Z}\\ l\in\mathbb{Z}}} 2\pi il(u_{k,l}) = \sum_{\substack{k\in\mathbb{Z}\\ l\in\mathbb{Z}}} -4\pi^2k^2(u_{k,l}) - \sum_{\substack{k\in\mathbb{Z}\\ l\in\mathbb{Z}}} u_{k,l} \qquad (3.6)$$

$$\Rightarrow u_t(x,t)_k + (4\pi^2k^2+1)u(x,t)_k = 0 \qquad (3.7)$$

$$u(x,t)_k = C(x)e^{-t(4\pi^2k^2+1)} = f(x)e^{-t(4\pi^2k^2+1)} \qquad (3.8)$$

$$\Rightarrow u(x,1)_k = f(x)e^{-(4\pi^2k^2+1)} \qquad (3.9)$$

$$\Rightarrow u(x,1)_n = f_n e^{-(4\pi^2n^2+1)} \qquad (3.10)$$

4

4.a

4.a.i

$$
\begin{aligned}
\int_{-\infty}^{\infty} \frac{1}{m^2+x^2}dx &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{m}{\cos^2(x)}\frac{1}{m^4}\frac{1}{\left(\frac{\cos^2(x)+\sin^2(x)}{\cos^2(x)}\right)^2}dx \\
&= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos^2(x)}{m^4}dx \qquad (4.1)\\
&= \frac{1}{m^3}\left[\frac{1}{2}x + \sin(2x)\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{\pi}{2m^3}
\end{aligned}
$$

4.a.ii

$$\begin{aligned}\int_{-\infty}^{\infty}\frac{1}{m^2+x^2}dx &= \frac{2\pi}{4m^2}\int_{-\infty}^{\infty}e^{-2m|x|}dx \\ &= \frac{\pi}{2m^2}\left(\left[\frac{e^{2mx}}{2m}\right]_{-\infty}^{0}+\left[\frac{e^{-2mx}}{-2m}\right]_{0}^{\infty}\right) \\ &= \frac{\pi}{2m^2}\left(\frac{1}{2m}+\frac{1}{2m}\right)=\frac{\pi}{2m^3}\end{aligned} \tag{4.2}$$

4.b

$$(a-|x|)_+ \quad \Rightarrow \quad \frac{2(1-\cos(ak))}{k^2} \tag{4.3}$$

$a=2$

$$(2-|x|)_+ \quad \Rightarrow 2\frac{1-\cos(2k)}{k^2}=\frac{4\sin^2(k)}{k^2} \tag{4.4}$$

$$e^{-|x|} \quad \Rightarrow \frac{2}{1+k^2} \tag{4.5}$$

$$\begin{aligned}\int_{-\infty}^{\infty}\frac{\sin^2(x)}{x^2(1+x^2)} &= \frac{1}{8}\int_{-\infty}^{\infty}\frac{4\sin^2(x)}{x^2}\frac{2}{1+x^2}dx \\ &= 2\pi\frac{1}{8}\int_{-\infty}^{\infty}e^{-|x|}(2-|x|)_+dx \\ &= \frac{\pi}{2}\int_0^2 e^{-x}(2-x)dx \\ &= \frac{\pi}{2}[(x-2)e^{-x}]_0^2-\int_0^2 e^{-x}dx \\ &= \frac{\pi}{2}(2+e^{-2}-1)=\frac{\pi(1+e^{-2})}{2}\end{aligned} \tag{4.6}$$

5

5.a

Ich sehe nicht wirklich wie das mit MMP zu tunf hat?

Let $u(r,\theta,\varphi)=f\circ K$

$$\nabla u=\frac{v_r}{||v_r||^2}\partial_r u+\frac{v_\theta}{||v_\theta||^2}\partial_\theta u+\frac{v_\varphi}{||v_\varphi||^2}\partial_\varphi u \tag{5.1}$$

$$v_r=\begin{pmatrix}\sin(\theta)\cos(\varphi)\\ \sin(\theta)\sin(\varphi)\\ \cos(\theta)\end{pmatrix}, \qquad ||v_r||^2=1 \tag{5.2}$$

$$\Rightarrow \frac{v_r}{||v_r||^2}=e_r \tag{5.3}$$

$$v_\theta=r\begin{pmatrix}\cos(\theta)\cos(\varphi)\\ \cos(\theta)\sin(\varphi)\\ -\sin(\theta)\end{pmatrix}, \qquad ||v_\theta||^2=r^2 \tag{5.4}$$

$$\Rightarrow \frac{v_\theta}{||v_\theta||^2}=\frac{e_\theta}{r} \tag{5.5}$$

$$v_\varphi=r\begin{pmatrix}-\sin(\theta)\sin(\varphi)\\ \sin(\theta)\cos(\varphi)\\ 0\end{pmatrix}, \qquad ||v_\varphi||^2=r^2\sin^2(\theta) \tag{5.6}$$

$$\frac{v_\varphi}{||v_\varphi||^2}=\frac{e_\varphi}{r\sin(\theta)} \tag{5.7}$$

$$\Rightarrow \nabla u = \partial_r u + \frac{1}{r}\partial_\theta u + \frac{1}{r\sin(\theta)}\partial_\varphi u \tag{5.8}$$

Irgendeine Formel aus dem Internet

$$(\mathrm{div}(A)) \circ K = \frac{1}{|\det(J)|} = \mathrm{div}(|\det(J)|G^{-1}(A \circ K)) \tag{5.9}$$

$$(JJ^{-1}))^{-1} = G^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{r} & 0 \\ 0 & 0 & \frac{1}{r\sin(\theta)} \end{pmatrix} \tag{5.10}$$

$$\Delta A = \mathrm{div}(\nabla A) \tag{5.11}$$

$$|\det(J)| = r^2 \sin(\theta) \tag{5.12}$$

$$\begin{aligned} \mathrm{div}(\nabla u) &= \frac{1}{r^2}\frac{\sin(\theta)}{\sin(\theta)}\partial_r(r^2\partial_r u) + \frac{1}{\sin(\theta)}\partial_\theta\left(\frac{1}{r}\frac{1}{r}\sin(\theta)\partial_\theta u\right) + \partial_\varphi\left(\frac{1}{r^2\sin^2(\theta)}\partial_\varphi u\right) \\ &= \partial_r^2 u + \frac{2}{r}\partial_r + \frac{1}{r^2}\frac{1}{\sin(\theta)}\partial_\theta(\sin(\theta)\partial_\theta) + \frac{1}{r^2\sin^2(\theta)}\partial_\varphi^2 \\ &= \left(\partial_r^2 + \frac{2}{r}\partial_r + \frac{1}{r^2}\Delta_{S^2}\right)(f \circ K) \end{aligned} \tag{5.13}$$

$$\Delta_{S^2} = \frac{1}{\sin(\theta)\partial_\theta}(\sin(\theta)\partial_\theta) + \frac{1}{\sin^2(\theta)}\partial_\varphi^2 \tag{5.14}$$

5.b

Rotationsinvariant

$$\Rightarrow \Delta_{S^2} = 0 \tag{5.15}$$

$$\Rightarrow (\Delta f) \circ K = \left(\partial_r^2 + \frac{2}{r}\partial_r\right)(f \circ K) \tag{5.16}$$

6

6.a

$$\hat{u}(x,k) = \int_{\mathbb{R}} u(x,y)e^{-iky}dx \tag{6.1}$$

$$\widehat{\partial_y^2 u}(x,k) = i^2k^2\hat{u}(x,k) \tag{6.2}$$

$$\Delta u(x,y) = 0 \Rightarrow \Delta\hat{u} = 0 \tag{6.3}$$

$$\begin{aligned} &\widehat{\partial_x^2 u}(x,k) + \widehat{\partial_y^2 u}(x,k) = 0 \\ \Leftrightarrow\ &\partial_x^2\hat{u}(x,k) - k^2\hat{u}(x,k) = 0 \end{aligned} \tag{6.4}$$

$$\lambda_{1,2} = \pm k \tag{6.5}$$

$$\hat{u}(x,k) = A(k)e^{kx} + B(k)e^{-kx} \tag{6.6}$$

6.b

$$\begin{aligned} u(0,y) = f_0(y) \Rightarrow \hat{u}(0,k) = \hat{f}_0(k) \\ u(1,y) = f_1(y) \Rightarrow \hat{u}(1,k) = \hat{f}_1(k) \end{aligned} \tag{6.7}$$

$$\begin{aligned} \hat{u}(0,k) = A(k) + B(k) = \hat{f}_0(k) \\ \hat{u}(1,k) = A(k)e^k + B(k)e^{-k} = \hat{f}_1(k) \end{aligned} \tag{6.8}$$

$$
\begin{aligned}
A &= \hat{f}_0 - B \\
\Leftrightarrow \big(\hat{f}_0 - B\big)e^k + Be^{-k} &= \hat{f}_1 \\
\Leftrightarrow \big(\hat{f}_1 - \hat{f}_0 e^k\big) &= B\big(e^{-k} - e^k\big) \\
\Rightarrow B(k) &= \frac{\hat{f}_1(k) - \hat{f}_0(k)e^k}{e^{-k} - e^k}
\end{aligned} \tag{6.9}
$$

$$
\begin{aligned}
A(k) &= \hat{f}_0 - \frac{\hat{f}_1(k) - \hat{f}_0(k)e^k}{e^{-k} - e^k} \\
&= \frac{\hat{f}_1(k) - \hat{f}_0(k)e^{-k}}{e^k - e^{-k}}
\end{aligned} \tag{6.10}
$$

6.c

$$
\begin{aligned}
\lim_{x\to 0} \left|\left|u(x,y)^2 - f_0(y)\right|\right|_2 &= \lim_{x\to 0} \int_{-\infty}^{\infty} |u(x,y) - f_0(y)|^2 dy \\
&= \lim_{x\to 0} \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \hat{u}(x,k) - \frac{\hat{f}_0(k)e^k - \hat{f}_1(k) + \hat{f}_1(k) - \hat{f}_0(k)e^{-k}}{e^k - e^{-k}} \right|^2 dk \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} \lim_{x\to 0} \frac{\left|\big(\hat{f}_1 - \hat{f}_0 e^{-k}\big)e^{kx} + \big(\hat{f}_0 e^k - \hat{f}_1\big)e^{-kx} - \hat{f}_0 e^k + \hat{f}_0 e^{-k}\right|^2}{|e^k - e^{-k}|^2} dy \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{|e^k - e^{-k}|^2} \left|\hat{f}_1 - \hat{f}_1 + \big(-\hat{f}_0 + \hat{f}_0\big)e^{-k} + \big(\hat{f}_0 - \hat{f}_0\big)e^{-k}\right|^2 dk = 0
\end{aligned} \tag{6.11}
$$